Università degli Studi di Napoli Federico II

Eclipse MOSAIC Co-Simulation Framework

(Integration of Traffic Simulation, Communication Networks, and C-ITS Applications)

Technical Learning Report

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Presentation Overview

01 **Project Objectives**
Learning goals and three-phase experimental framework

02 **MOSAIC**
Federate integration and co-simulation framework

03 **Phase 1: Traffic Simulation**
Parameter manipulation and congestion dynamics

03 **Phase 2: C-ITS Communication**
V2X experiments and penetration rate analysis

04 **Phase 3: RSU and Message Propagation**
Infrastructure-to-vehicle

05 **Skills & Status**
Competencies acquired and future directions

Project Objectives & Learning Framework

Primary Objective

Following your guidance, we embarked on a collaborative technical learning journey to master the Eclipse MOSAIC co-simulation environment. This foundational phase was designed to build expertise in cooperative intelligent transport systems (C-ITS) through hands-on experimentation.

Three-Phase Approach

01 Traffic Simulation

Parameter manipulation and congestion dynamics

02 C-ITS Communication

V2X messaging and penetration analysis

03 RSU Integration

Infrastructure-based traffic control

• MOSAIC

• Core

SUMO Federate

Traffic

Microscopic traffic simulation handling vehicle dynamics, lane-changing, and route assignment

SNS (Communication)

Network

Simulates V2X communication networks including signal propagation, interference, and message routing

Application Federate

Logic

Hosts C-ITS applications like CAM generation, speed advisory, and vehicle behavior control

• Federate Interaction Flow

- **SUMO**
  Vehicle positions & traffic state
- **SNS**
  Message transmission & reception
- **Application**
  C-ITS logic & behavior control
- **SUMO (Feedback)**
  Updated vehicle behavior

Traffic Simulation

HelloWorld Scenario Execution

We initiated our learning journey with the HelloWorld scenario to understand the fundamental integration between MOSAIC components. Using the real-time visualization mode with the `-b 1` flag, we observed live traffic flow, communication networks, and application layer activations simultaneously.

```
Command: mosaic.bat -s HelloWorld -v -b 1
```

Parameter Modifications

Simulation Duration scenario_config.json	100s	**200s**
Maximum Speed Vehicle type parameter	70 km/h	**10 km/h**
Target Flow Traffic demand	4000 veh/h	**5000 veh/h**
Vehicle Count	50	**500**

Observed Traffic Dynamics

Speed-Capacity Relationship

Reducing maxSpeed from 70 to 10 km/h demonstrated the fundamental traffic engineering principle: **lower speed directly reduce roadway capacity**, creating bottlenecks even with moderate demand.

Congestion Formation

Increasing targetFlow to 5000 veh/h with 500 vehicles created **visible queue formation and shockwave propagation**. We observed how high inflow exceeds capacity, leading to sustained congestion.

Queue Dynamics

Visual verification through the MOSAIC visualizer confirmed **queue formation at bottlenecks and gradual dissipation** when demand decreased, validating SUMO's microscopic behavior models.

C-ITS Communication

Comparative Experiment Structure

Building on Phase 1's traffic fundamentals, we designed a controlled experiment to isolate the effects of V2X communication on traffic flow dynamics. The experiment compared baseline behavior against C-ITS-equipped scenarios.

A — Baseline Scenario

- No V2X application
- No CAM generation
- Standard human reaction

B — C-ITS Scenario

- VehicleCamSendingApp
- CAM message generation
- V2V communication

Experimental Parameters

Duration	Application	Interaction
200s	CAM App	V2V

Cooperative Awareness Messages

What is CAM?

Cooperative Awareness Messages are **periodic V2X broadcasts** containing vehicle position, speed, heading, and acceleration data, enabling surrounding vehicles to maintain situational awareness.

Application Layer Role

The **VehicleCamSendingApp** generates CAM messages at regular intervals, which are transmitted through SNS to nearby vehicles, triggering behavioral adaptations.

SUMO Behavior Influence

Received CAM messages enable vehicles to **anticipate traffic conditions ahead**, adjusting speed and following distance proactively rather than reactively.

Penetration Rate & Reaction Time Analysis

Three Penetration Scenarios

To understand the progressive impact of C-ITS deployment, we tested three penetration scenarios with varying tau (reaction time) values, representing different levels of market penetration for connected vehicles.

0% No C-ITS

tau = 1.5s (all vehicles)

Standard human reaction time, no communication capability

50% Mixed Fleet

tau = 0.5s (50%) + **tau = 1.5s** (50%)

Half connected vehicles, half conventional vehicles

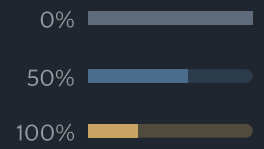
100% Full C-ITS

tau = 0.5s (all vehicles)

All vehicles equipped with V2X, reduced reaction time

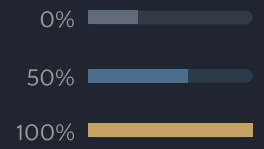
Performance Comparison

Inter-Vehicle Gap



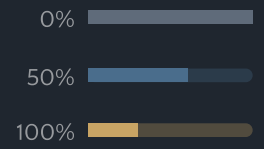
↓ Higher penetration reduces gaps significantly

Queue Dissipation Speed



↑ Higher penetration enables faster recovery

Shockwave Intensity



↓ Higher penetration weakens shockwaves

RSU Infrastructure & Speed Advisory

Roadside Unit as Traffic Control Center

To advance beyond vehicle-to-vehicle communication, we introduced infrastructure-to-vehicle (I2V) communication through Roadside Units (RSUs). The RSU acts as a **road traffic control center**, broadcasting authoritative messages that all vehicles receive simultaneously.

Infrastructure-to-Vehicle Communication

The RSU transmits **real-time speed advisory messages** based on traffic conditions ahead. Unlike V2V communication where information propagates gradually, I2V enables **synchronized vehicle reactions** across the entire network.

Independence from Traffic Light Phases

With RSU-based control, we are **no longer dependent on fixed traffic light timing**. The system becomes dynamically controllable based on real-time traffic conditions, enabling adaptive traffic management strategies.

Message Flow

RSU (Infrastructure)

Generates speed advisory messages based on traffic conditions

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SNS (Communication)

Transmits messages to all vehicles in broadcast range

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Vehicles (Reaction)

Receive and adapt speed simultaneously, creating coordinated behavior

Real-Time

Messages reflect current conditions

Synchronized

All vehicles react simultaneously

AdvisoryCar Implementation

AdvisoryCar Configuration

We implemented the `advisedcar` configuration in the `mapping_config.json` file to enable RSU-based speed advisory functionality. This specialized vehicle type receives and reacts to infrastructure messages.

Configuration Parameters

- **Vehicle Type:** advisedcar with V2X receiver capability
- **Application:** Speed advisory message handler
- **Communication:** I2V message reception via SNS
- **Behavior:** Dynamic speed adaptation based on RSU instructions

Temporal Behavior Observation

0-40s Baseline Period

During the initial 40 seconds, vehicles operate under **normal conditions** without RSU intervention. This establishes the baseline traffic flow pattern and vehicle behavior for comparison.

40s+ RSU Activation Period

After the 40-second, the RSU begins broadcasting **speed reduction messages**. New vehicles entering the network receive these messages and **adapt their behavior accordingly**, demonstrating the infrastructure control capability.

Observed Phenomena

- **Speed Adaptation**

New entering vehicles **move slowly** after receiving RSU speed reduction messages, creating a controlled flow pattern that prevents congestion buildup.

- **Coordinated Behavior**

Multiple vehicles react **synchronously** to the same infrastructure message, demonstrating the power of centralized traffic control through V2X communication.

- **Integration Validation**

The experiment successfully validated that **Application + SNS + SUMO** work together seamlessly to enable infrastructure-based traffic management.

• Technical Skills & Competencies Acquired

Simulation Configuration

- **Autonomous Execution:** Configure and run MOSAIC simulations independently
- **Scenario Management:** Understand folder structure and configuration files
- **Runtime Control:** Execute simulations with various flags and parameters

Traffic Simulation

- **Parameter Manipulation:** Modify traffic flow, speed, and vehicle counts
- **Congestion Analysis:** Understand queue formation and shockwave dynamics
- **Behavioral Modeling:** Configure vehicle types and reaction parameters

V2X Application

- **App Deployment:** Equip vehicles with C-ITS applications
- **CAM Generation:** Implement cooperative awareness messaging
- **Interaction Mechanisms:** Understand how federates communicate

Thank you for your Attention